DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY

PROJECT REPORT

Vehicle Detection System

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1. ABSTRACT

Traffic congestion and inefficient road monitoring have become major challenges in modern urban areas. Traditional traffic observation methods require manual effort and often produce inaccurate results. To address this issue, this project presents an AI-based Vehicle Detection System using the YOLOv8 deep learning model and computer vision techniques.The system is designed to automatically detect and count different types of vehicles, including cars, buses, trucks, and motorcycles, from traffic images. The uploaded image first undergoes preprocessing techniques such as contrast enhancement and noise reduction to improve image quality. After preprocessing, the YOLOv8x pretrained object detection model identifies vehicles and generates bounding boxes with confidence scores. To improve counting accuracy, overlapping detections are removed using an IoU-based duplicate removal technique.

The final system provides visual detection results along with total vehicle counts, making traffic monitoring faster, more accurate, and efficient. This project demonstrates the practical application of artificial intelligence in intelligent transportation systems and highlights the potential of deep learning for future smart city traffic management solutions.

2. INTRODUCTION

Traffic management has become a major challenge in modern cities. Traditional manual methods of vehicle counting are time-consuming and error-prone with the help of **Artificial Intelligence** and **Computer Vision**, this process can be automated for image-based traffic analysis..

In this project, we use **YOLOv8 (You Only Look Once version 8)** to detect and classify vehicles in images. The system processes input images, detects vehicles, and provides accurate counts for each category.

3. PROBLEM STATEMENT

Manual traffic monitoring faces several issues:

- Time-consuming process
- Human errors in counting
- Lack of real-time analysis
- Inefficient traffic density evaluation

Therefore, an automated system is required to improve accuracy and speed.

4. OBJECTIVES

- Detect vehicles in traffic images
- Classify vehicles into categories (Car, Bus, Truck, Motorcycle)
- Count total vehicles accurately
- Improve traffic monitoring efficiency
- Apply deep learning in real-world scenarios

5. LIBRARIES USED

Python	Programming language
OpenCV	Image processing and visualization
YOLOv8	Object detection model
NumPy	Numerical operations
Google Colab	Development environment

6. METHODOLOGY

The system follows these steps:

1. Image upload by user
2. Image preprocessing (enhancement & noise removal)
3. Vehicle detection using YOLOv8
4. Bounding box generation
5. Duplicate removal using IoU + NMS
6. Vehicle counting
7. Display of final results

Figure 1: Original Traffic Image Used for Vehicle Detection

7. IMAGE PREPROCESSING

Image preprocessing improves detection accuracy using:

- CLAHE (Contrast Enhancement)
- LAB color space conversion
- Noise reduction using denoising filter
- This helps improve image quality before detection.

Figure 2: Preprocessed Image after Enhancement

8. VEHICLE DETECTION USING YOLOv8

The system uses **YOLOv8x pretrained model**, which is a powerful deep learning model for object detection.

It detects:

- Car
- Bus

- Truck
- Motorcycle Each detection

includes:

- Bounding box
- Class label
- Confidence score

Figure 3: Vehicle Detection Output using YOLOv8

9. DUPLICATE REMOVAL (IOU + NMS)

To improve accuracy, duplicate detections are removed using:

- **IOU (Intersection over Union)**
- **Non-Maximum Suppression (NMS)**

This ensures that each vehicle is counted only once.

10. RESULTS

The system successfully:

- Detects multiple vehicles in a single image
- Classifies vehicle types correctly
- Provides counting

Figure 4: Vehicle Counting Result Display

11. PERFORMANCE ANALYSIS

The system shows strong performance in:

- Daytime traffic images
- Multiple vehicle detection
- Fast processing speed
- High accuracy detection

12. LIMITATIONS

- Performance decreases in low light conditions
- Small distant vehicles may be missed
- Overlapping vehicles can reduce accuracy

13. FUTURE ENHANCEMENTS

- Real-time CCTV integration
- Traffic density prediction system
- Vehicle speed detection
- Smart city traffic management system
- Use of advanced models like YOLOv9 or DeepSORT

14. CONCLUSION

This project successfully developed an intelligent vehicle detection and counting system using the YOLOv8 deep learning model and computer vision techniques. The system is capable of accurately detecting multiple vehicle categories, including cars, buses, trucks, and motorcycles, from traffic images with high efficiency and speed.

By integrating image preprocessing, AI-based object detection, and IoU-based duplicate removal, the proposed system improves detection accuracy and reduces false counting. The generated output provides clear visual results with labeled bounding boxes and total vehicle statistics, making the system suitable for modern traffic monitoring applications.

The project demonstrates how artificial intelligence can replace traditional manual traffic observation methods with smarter and more reliable solutions. It also highlights the growing importance of deep learning in intelligent transportation systems, smart city development, and real-time traffic management. Overall, the system achieved its objectives successfully and proved to be an effective, practical, and scalable solution for automated vehicle detection.